CS3002E Compiler Design

Monsoon 2025

Lecture #19

Bottom-Up Parsing- 8: Miscellaneous topics

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Grammar with ϵ productions - LR(0) items

- For a production $A \rightarrow \epsilon$ add item $A \rightarrow .$
- Item $A \rightarrow .$ indicates a reduction by $A \rightarrow \epsilon$
  - push A to stack without popping anything

Example

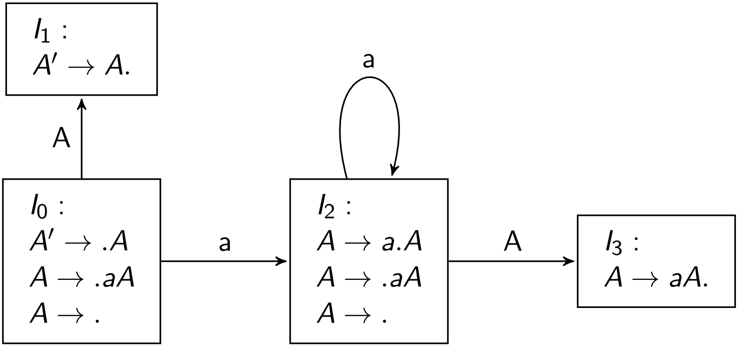
$A' \rightarrow A \qquad\qquad A \rightarrow aA \mid \epsilon$

Initial State $=\{A' \rightarrow .A,\ A \rightarrow .aA,\ A \rightarrow .\}$

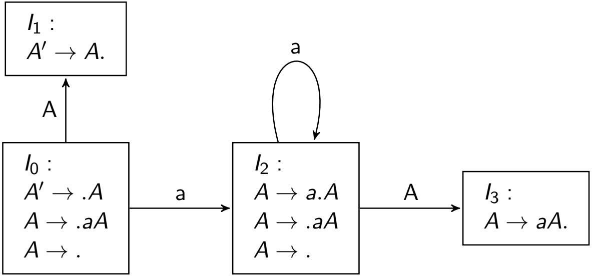
Exercise:

1. Check if the grammar is LR(0). If not, is it SLR / LR(1)?
2. What are the state transitions of an LR parser for the inputs ϵ, a, aa and aaa?

LR(0) Automaton



SLR Parsing



What are the state transitions for the inputs ϵ, a, aa, aaa?

Shift-Reduce Parser: Configuration

A *configuration* is a pair $(X_1X_2 \ldots X_m,\ a_ia_{i+1} \ldots a_n)$, where $X_1,\ X_2, \ldots X_m$ are the symbols in the stack (from bottom to top) and $a_i,\ a_{i+1} \ldots a_n$ are the input symbols remaining to be read.

Parser configurations ?

$$E \rightarrow \; E + id \mid id$$

$$E \underset{rm}{\Rightarrow} E + id \underset{rm}{\Rightarrow} id + id$$

Stack	Input	Action
$\$$	$id + id\$$	shift
$\$id$	$+id\$$	reduce using $E \rightarrow id$
$\$E$	$+id\$$	shift
$\$E+$	$id\$$	shift
$\$E + id$	$\$$	reduce using $E \rightarrow \; E + id$
$\$E$	$\$$	accept

Shift-Reduce Parsing

$$E \underset{rm}{\Rightarrow} E + id \underset{rm}{\Rightarrow} id + id$$

Stack	Input	Action
$\$$	$id + id\$$	shift
$\$id$	$+id\$$	reduce using $E \rightarrow id$
$\$E$	$+id\$$	shift
$\$E+$	$id\$$	shift
$\$E + id$	$\$$	reduce using $E \rightarrow \; E + id$
$\$E$	$\$$	accept

- The stack contents concatenated with the remaining input gives a right-sentential form.

Shift-Reduce Parsing: Viable Prefix

$$E \underset{rm}{\Rightarrow} E + id \underset{rm}{\Rightarrow} id + id$$

Stack	Input	Action
$\$$	$id + id\$$	shift
$\$id$	$+id\$$	reduce using $E \rightarrow id$
$\$E$	$+id\$$	shift
$\$E+$	$id\$$	shift
$\$E + id$	$\$$	reduce using $E \rightarrow \ E + id$
$\$E$	$\$$	accept

- The stack contents concatenated with the remaining input gives a right-sentential form.
- Stack contains a prefix of a right-sentential form - known as Viable Prefix .

Viable Prefix

- A prefix of a right sentential form that can appear on the stack of a shift-reduce parser.
- The prefix does not extend past the right end of the rightmost handle.
- Not all prefixes of right sentential forms can be viable prefixes.
    - when a handle appears in the stack, the parser reduces without shifting any more symbols.

Viable Prefix: Example

$$E \underset{rm}{\Rightarrow} E + id \underset{rm}{\Rightarrow} id + id$$

Stack	Input	Action
$\$$	$id + id\$$	shift
$\$id$	$+id\$$	reduce using $E \rightarrow id$
$\$E$	$+id\$$	shift
$\$E+$	$id\$$	shift
$\$E + id$	$\$$	reduce using $E \rightarrow \; E + id$
$\$E$	$\$$	accept

For the sentential form $id + id$

ϵ and id are the viable prefixes.

$id+$ is not a viable prefix.

Viable Prefix: Exercise

$$E \rightarrow \; E + id \mid id$$

$$E \underset{rm}{\Rightarrow} E + id \underset{rm}{\Rightarrow} id + id$$

Stack	Input	Action
$\$$	$id + id\$$	shift
$\$id$	$+id\$$	reduce using $E \rightarrow id$
$\$E$	$+id\$$	shift
$\$E+$	$id\$$	shift
$\$E + id$	$\$$	reduce using $E \rightarrow \; E + id$
$\$E$	$\$$	accept

Exercise: List the viable prefixes of $E + id$

Valid items

- An item $A \rightarrow \beta_1.\beta_2$ is *valid* for a viable prefix $\alpha\beta_1$ if there is a derivation $S' \overset{*}{\underset{rm}{\Rightarrow}} \alpha A w \underset{rm}{\Rightarrow} \alpha\beta_1\beta_2 w$
- Set of valid items for a viable prefix γ is the set of items reached from the initial state along the path labeled γ in the LR(0) automaton for the grammar.

References

References:

- Aho A.V., Lam M.S., Sethi R., and Ullman J.D. Compilers: Principles, Techniques, and Tools (ALSU). Pearson Education, 2007.

Further reading:

- ALSU sections 4.5.2, 4.5.3, 4.6.5